

A herbal antifungal formulation of *Thymus serpyllum*, *Origanum vulgare* and *Rosmarinus officinalis* for treating ovine dermatophytosis due to *Trichophyton mentagrophytes*

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Summary

A number of herbal products with anti-inflammatory, antiseptic and antimycotic properties are available for dermatological usage. The successful treatment of 13 sheep affected by ringworm due to *Trichophyton mentagrophytes* with a mixture consisting of essential oils (EOs) of *Thymus serpyllum* 2%, *Origanum vulgare* 5% and *Rosmarinus officinalis* 5% in sweet almond (*Prunus dulcis*) oil. The effectiveness of EOs and of the major components of the mixture (thymol, carvacrol, 1,8 cineole, α -pinene, p-cymene, γ -terpinene) against the fungal clinical isolate was evaluated by a microdilution test. Thirteen animals were topically administered with the mixture twice daily for 15 days. The other sheep were administered with a conventional treatment (seven animals) or left untreated (two animals). Minimum inhibitory concentration (MIC) values were 0.1% for *T. serpyllum*, 0.5% for *O. vulgare*, 2.5% for *I. verum* and 5% for both *R. officinalis* and *C. limon*. Thymol and carvacrol showed MICs of 0.125% and 0.0625%. A clinical and aetiological cure was obtained at the end of each treatment regimen in only the treated animals. Specific antimycotic drugs licenced for food-producing sheep are not available within the European Community. The mixture tested here appeared to be a versatile tool for limiting fungal growth.

Key words: Essential oils, *Trichophyton mentagrophytes*, *Thymus serpyllum*, *Origanum vulgare*, carvacrol, thymol.

Introduction

Essential oils (EOs) are very complex natural mixtures, which are formed by aromatic plants as secondary metabolites. These compounds are extracted with various methods and can contain about 20–60 components with different concentrations. EOs are characterised by a few major components, whose high concentrations determine their biological properties.¹ Due to their antimicrobial activity, pharmaceutical use

of these compounds has become increasingly widespread as an alternative to synthetic chemical products¹ or in combination therapies.² The antifungal activity of some EOs has been assessed both *in vitro*³ and *in vivo*.⁴ The demand for natural products is increasing in veterinary medicine also, particularly considering the perspective of animal welfare, and friendly breeding, and owners often welcome this treatment.

Dermatophytosis (or ringworm) is a skin disease caused by keratinophilic fungi belonging to the anamorph genera *Microsporum*, *Trichophyton* and *Epidermophyton*, which colonise the superficial layer of the stratum corneum. Such infections in sheep are generally regarded as uncommon.⁵ Clinical signs are usually mild and active lesions heal after 6–8 weeks. However, fungal elements usually spread in the environment, and difficulties in managing and controlling ringworm can arise, especially considering that the specific

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Submitted for publication 6 June 2012

Revised 21 November 2012

Accepted for publication 28 November 2012

antimycotic treatment for sheep and other food producer animals is banned in European Community. Furthermore, when a zoonotic mycotic species is involved, dermatophytoses have an impact on public health.

Trichophyton verrucosum is the main fungal species involved in ovine ringworm, followed by *Microsporum canis* and to a lesser extent *M. gypseum*.⁶ Infections caused by *T. mentagrophytes* have been less frequently encountered.⁵

This study reports the occurrence of ovine ringworm associated with *T. mentagrophytes*. The sensitivity pattern of fungal isolate to some EOs and their major components is determined. The clinical and aetiological resolution of the disease after the topical administration of a herbal formulation of *Thymus serpyllum*, *Origanum vulgare* and *Rosmarinus officinalis* is also described.

Materials and methods

An outbreak of ovine ringworm occurred in a Zerasca sheep flock, consisting of approximately 200 animals, kept in northern Tuscany (Central Italy). The subjects were divided into three separate shelters used for winter housing. Twenty-two ewes from a group of 60 adults simultaneously exhibited several circular facial lesions, with a diameter of up to 6 cm. The lesions were alopecic and erythematous, sometimes covered by crusts (Figure). The fleece was not affected by the lesions. A clinical score was assigned to the diseased animals according to Westhoff *et al.*⁷ A score of 2 was given to sheep showing hairless areas and/or areas of thinning hair with local inflammation such as erythema and swelling, and score of 3 to sheep showing hairless areas with mild crusts, scales and/or exudates. Thirteen sheep had a score of 2, and the rest 3. The animals were kept in a shelter that had previously been used by other livestock, but had then been converted into housing for sheep. Hair samples were obtained by brushing the affected areas after cleaning with absolute ethanol to avoid the growth of environmental contaminant moulds. The hair specimens were seeded onto Sabouraud Dextrose Agar (SDA), mixed with 0.05% cycloheximide and the plates were incubated at 25 °C for 10 days. The cultures were checked for mycotic growth from day 4 postinoculation. All cultured samples yielded a pure growth of *T. mentagrophytes* var. *mentagrophytes*, identifiable by macro- and microscopic features.

Considering the contagiousness and zoonotic potential of this dermatophyte, treatment was strongly advised. Specific antimycotic drugs licenced for food-producing sheep are not available within the European

Community, and Iodine 5% is now rarely used⁸ as it can stain the hair. Consequently, an unconventional treatment using herbal products was selected. EOs from *O. vulgare* and *T. serpyllum* were chosen for their antidermatophyte activity.^{9,10} *R. officinalis* was added due to its marked anti-inflammatory properties.¹¹

The subjects were randomly assigned to each group. Thirteen animals (eight with a score of 2) were topically administered a mixture made up of *O. vulgare* (5%), *R. officinalis* (5%) and *T. serpyllum* (2%) in sweet almond oil, twice a day for 15 days directly on the affected areas. The other six sheep (seven with a score of 2) were isolated, administered with a conventional treatment (iodine 5%) daily for 5 days, then weekly for three treatments (Hopkins F. and Gill W. Ringworm (club lamb fungus) in sheep, Agricultural Extension Service, The University of Tennessee) and were kept as controls. Two untreated animals (scores of 2 and 3), used to evaluate the clinical outcome of the infection, were sheltered separately. Fungal isolates were used to perform an *in vitro* sensitivity assay using the following EOs: *Citrus limon* s. 90660, *Illicium verum* s. 70487, *O. vulgare* s. 60326, *R. officinalis* s. 60734 and *T. serpyllum* s. 90017. *C. limon* and *I. verum* were included to verify their potential antidermatophyte activity.^{12,13}

Volatile constituents of each EO were analysed by GC-MS with a CP-3800 gas chromatograph equipped with HP-5 capillary column (30 m × 0.25 mm; coating thickness, 0.25 mm) and Varian Saturn 2000 ion trap mass detector. Analytical conditions were as follows: injector and transfer line temperature, 220 and 240 °C, respectively; oven temperature, programmed from 60 to 240 °C at 3 °C per min; carrier gas, helium at 1 ml min⁻¹; injection, 0.2 ml (10% hexane solution); split ratio, 1:30. The constituents were identified by comparing the retention time with the authentic samples, on the basis of their linear indices relative to a series of *n*-alkanes (C8-C23). Further identifications were carried out using a homemade library of mass spectra created from pure substances and components of known oils, and the MS literature data (NIST 2000, ADAMS).

The *in vitro* antimycotic activity of EOs was evaluated on the clinical isolate of *T. mentagrophytes*, by means of a microdilution test carried out as previously described using a semisolid malt extract medium (MEA) with 1% agar.¹⁴

Portions of approximately 1 mm³ non-sporulating mycelia from SDA were used as fungal inocula in 24 wells plates (PBI International, Milan, Italy). EOs were diluted in sweet almond oil to obtain a 20% stock solution. Serial dilutions in semisolid MEA were made to achieve final concentrations ranging from 0.01% to

10%. To avoid bias due to differing amounts of sweet almond oil, each test was carried out using a concentration equal to 40% of this oil. This concentration was also added to the medium in control cultures. Further controls with medium alone and a synthetic antimycotic drug (enilconazole) were carried out. The cultures were incubated at 25 °C until the full development of mycotic growth in control wells was observed. Portions of the inocula where growth was not visually noticed were removed, twice washed in sterile saline and then seeded onto MEA plates to evaluate the viability of the fungi. All tests were performed in quadruplicate. To evaluate the antimycotic activity of the major components of the mixture, a clinical isolate was incubated with α pinene (from 0.03% to 10%), p-cymene (from 0.03% to 10%), 1,8 cineole (0.03% to 10%), γ terpinene (0.03% to 10%), thymol (0.03% to 10%) and carvacrol (0.03% to 10%), diluted in sweet almond oil, in semisolid MEA, as previously reported for EOs. All products were provided by Sigma-Aldrich (Milan, Italy). The viability of the inoculum was evaluated as described above.

Results and discussion

After each treatment, skin lesions gradually regressed and hair regrowth started. Adverse effects were not

observed. Control cultures performed 2 weeks after the end of the treatments were all negative, and no relapse of the disease was observed during a follow-up period of 6 weeks, whereas untreated subjects remained both symptomatic and culturally positive.

The chemical composition of the tested oils is reported in Tables 1 and 2. The main components were typical monoterpenes such as thymol (*O. vulgare*), carvacrol (*T. serpyllum*, *O. vulgare*), p-cymene (*O. vulgare*, *T. serpyllum*), γ -terpinene (*O. vulgare*, *T. serpyllum*, *C. limon*), limonene (*C. limon*, *R. officinalis* and *I. verum*), 1,8-cineole (*R. officinalis*), E-anethol (*I. verum*), α - and β -pinene (*C. limon*, *R. officinalis*).

Thymus serpyllum and *O. vulgare* showed the lowest minimum inhibitory concentrations (MICs), which were active at 0.1% and at 0.5%, respectively, followed by *I. verum* (2.5%), whereas *R. officinalis* and *C. limon* showed MIC at 5%. The overall MIC value of the mixture of *T. serpyllum* (2%), *O. vulgare* (5%) and *R. officinalis* (5%) was 20%. The MIC for enilconazole was 0.1%.

Minimum fungicidal concentrations (MFCs) values were 0.1% (*T. serpyllum*), 0.5% (*O. vulgare*) and 7.5% (*R. officinalis*, *C. limon*, *I. verum*). The MFC value of the mixture was 50%. The examined compounds showed different sensitivity patterns. Thymol and carvacrol were the most effective showing MIC values of 0.125% and 0.0625%, respectively. MFCs were higher than

Table 1 Chemical composition of the tested essential oils (area percentage, %).

	<i>Origanum vulgare</i>	<i>Thymus serpyllum</i>	<i>Citrus limon</i>	<i>Rosmarinus officinalis</i>	<i>Illicium verum</i>
Hydrocarbon monoterpenes	34.6	16.2	92.5	50.5	4.8
Oxygenated monoterpenes	58.1	80.3	4.9	46.1	95.1
Total monoterpenes	92.7	96.5	97.4	96.6	99.9
Hydrocarbons sesquiterpenes	5.8	2.2	2.6	2.3	
Oxygenated sesquiterpenes	0.2	0.4		0.2	
Total sesquiterpenes	6.0	2.6	2.6	2.5	
Others	1.0				

Table 2 Main Constituents of the tested essential oils (area percentage, %).

Compound	LRI ¹	<i>Origanum vulgare</i>	<i>Thymus serpyllum</i>	<i>Citrus limon</i>	<i>Rosmarinus officinalis</i>	<i>Illicium verum</i>
α -Pinene	939			2.3	23.4	0.7
β -Pinene	979			13.7	3.8	
p-Cymene	1026	14.3	5.2			
Limonene	1029			59.2	4.7	3.1
1,8-Cineole	1033				27.5	
γ -Terpinene	1060	11.2	4.7	10.8		
Linalool	1098	3.2	2.1			
Camphor	1146				7.1	
(E)-Anethol	1285					93.7
Thymol	1290	45.0				
Carvacrol	1298	4.0	72.0			

¹Linear Retention Index calculated on the basis of a DB-5 capillary column.



Figure 1 Circular, alopecic and erythematous lesions on the face of one affected animal.

Table 3 Minimal inhibiting concentration and minimal fungicide concentration of *Thymus serpyllum*, *Origanum vulgare* and *Rosmarinus officinalis* main constituents.

Compound	MIC (%)	MFC (%)
α -Pinene	8	>8
p-Cymene	>8	>8
1,8-Cineole	1	>8
γ -Terpinene	>12	>12
Thymol	0.125	1
Carvacrol	0.0625	4

expected, 1% for thymol and 4% for carvacrol. MICs/MFCs are detailed in Table 3.

The treatment proposed is based on EOs with a proven antimycotic activity, such as *T. serpyllum* and *O. vulgare*. These compounds have shown a specific activity against dermatophytes.^{9,14} The biological effects of EOs seem to act as prooxidants affecting inner cell membranes and organelles such as mitochondria.¹ High amounts of thymol and carvacrol were detected. It is hypothesised that these lipophilic compounds penetrate the cells and target the ergosterol biosynthesis pathway, thus, impairing its biosynthesis.¹⁵ Thymol, carvacrol, p-cymene and γ -terpinene have also been shown to possess considerable antifungal activities resulting from membrane damage.¹⁰

In this study MICs of single constituents were higher than those reported in the literature.^{10,16} These discrepancies could be explained by the fact that the fungi used in our study were maintained in more suitable conditions in terms of both the media (MEA rather than RPMI) and the diluents (sweet almond oil rather than DMSO). Almond oil seems to increase the MICs compared with DMSO, in fact the test method had a

significant influence on the measurement of antifungal activity, as been previously reported.¹⁴

Inherent differences between fungal species also influence the *in vitro* antifungal activity of EOs. In a similar trial carried out using a mixture of EOs in *M. canis* infected cats, the MFC value of the mixture was 0.05%,¹⁷ indicating a higher sensitivity of this dermatophyte compared with *T. mentagrophytes*. Pinto *et al.* and Vale-Silva *et al.*^{9,10} corroborate this finding on collection strains. However, the mixture has also yielded promising results *in vivo* in cats.¹⁷

There is little information in the literature regarding the occurrence of *T. mentagrophytes* as the causative agent of ovine ringworm. The mould is distributed worldwide, and small rodents are considered as a reservoir. Infections are widespread in animals and humans, and are transmitted through shared environments and equipment. In this study, the disease occurred in animals housed in an environment accessible by mice, rats and voles. Overcrowded housing, bad bedding conditions and the poor ventilation of shelters, where sheep congregate during the winter season could have facilitated the outbreak. The zoonotic character of the disease, the spread of infection to other animals, the lack of a fast self-cure and the risk of environmental contamination make treatment extremely advisable.

Even though the number of animals involved in this trial was small, topical treatment with the EOs was practical and effective and may be a useful tool in ovine species, considering that the present protocol showed a better result than conventional treatment with regard to the speed of resolution.

Acknowledgment

The authors would like to thank Flora (Lorenzana, Pisa, Italy) for providing the EOs samples used in the study.

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